LATTICE

Ledgered Agent Traces for Transparent, Inspectable Collaborative Execution

v1.2.1 · Content-Addressed Accountability Protocol for Multi-Agent AI Systems

LATTICE is a framework-agnostic Python library that makes every AI agent decision **traceable**, **tamper-evident**, and **auditable**. Every claim is content-addressed (SHA-256), cryptographically signed (Ed25519), and linked in a Directed Acyclic Graph from conclusions to raw evidence.

KEY CAPABILITIES

- **Backward Traceability** — Walk from any conclusion to its raw evidence
- **Tamper Detection** — Content-addressed claims + Ed25519 signatures
- **Effective Confidence** — Min-path propagation exposes inflated confidence
- **Revocation Waterfall** — Cascade invalidation through dependency graph
- **Runtime Monitoring** — Zero-friction `@lattice_monitor` decorator
- **Web Dashboard** — FastAPI + D3.js interactive DAG visualization
- **CLI** — `lattice trace|audit|verify|stats|export`

PERFORMANCE

0.65 ms Claim creation + signing	**66.9 ms** 10K-node DAG traversal
0.19 ms Instrumentation overhead	**$O(n^{1.03})$** Near-linear scaling

ARCHITECTURE

- **Storage:** SQLite (WAL mode) — single portable file
- **Crypto:** SHA-256 content addressing + Ed25519 signatures
- **Dependencies:** 3 core packages (cryptography, click, rich)
- **Offline-first:** No cloud, no external services
- **Framework-agnostic:** Works under LangGraph, CrewAI, AutoGen, or raw Python

CASE STUDY: OSINT PIPELINE

Three-agent investigation (Harvester → Analyzer → Reporter) completing in **1.30 ms**:

- 9 signed claims across 3 agents, backed by 5 evidence blobs
- 7 collection methods (nslookup, WHOIS, HTTP, crt.sh, passive DNS, LLM analysis, synthesis)
- Audit detected 1 inflated confidence: stated 0.75, effective 0.70
- All signatures cryptographically verified

TEST SUITE

- **89 tests** covering content addressing, signatures, DAG traversal, revocation, cycle detection, effective confidence, CLI, dashboard API, and runtime monitor
- CI via GitHub Actions on Python 3.10–3.13

APPLICATIONS

- **OSINT & Threat Intelligence** — Auditable investigation chains
- **Security Consulting** — Client-facing reports with provenance
- **EU AI Act Compliance** — Explainability via claim DAGs
- **Autonomous Systems** — Mission-critical decision accountability
- **Research Pipelines** — Reproducible multi-agent analysis

TECHNICAL SPECS

- Python 3.10+ · MIT License · ~2,400 LOC
- `pip install lattice-core`
- Full benchmark suite included

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